

Faculty of Mathematics and Informatics

Software Design Project - EuroGlobe Bank

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1. Introduction and Overview

1.1 Product Overview

EuroGlobe Bank is a next-generation financial services platform designed to modernize and unify fragmented banking experiences. Built to cater to both individual and institutional clients across the European Union, the platform combines modern software engineering practices with strict compliance frameworks to ensure robust, secure, and user-friendly digital banking solutions.

1.2 Core Problem Statement

Traditional retail and commercial banking infrastructure continues to be plagued by operational inefficiencies, opaque customer-facing processes, and restrictive product capabilities. EuroGlobe Bank directly addresses four critical industry pain points:

1. **Mortgage Transparency and Process Inefficiency:** Traditional mortgage applications suffer from slow approval cycles, lack of real-time visibility into application stages, and complex fixed/variable rate terms that confuse applicants. EuroGlobe Bank introduces a dynamic, state-machine-driven mortgage lifecycle that provides total visibility and automated workflow progression.
2. **Friction in Vehicle Leasing:** Car leasing processes traditionally require extensive manual documentation and fragmented third-party evaluations. EuroGlobe Bank streamlines vehicle leasing through automated eligibility calculations, customizable terms, and digital agreement handling.
3. **Inflexible Group Financial Control:** Standard joint bank accounts lack granular control mechanism, creating trust risks for shared apartments, joint ventures, or group travel. EuroGlobe Bank implements shared multi-signature vaults requiring configurable consent thresholds (e.g., 2-of-3 signature rules) before executing withdrawals or transfers.
4. **Passive and Ineffective Micro-Savings:** Conventional savings accounts fail to engage users effectively. EuroGlobe Bank integrates real-time transaction round-ups, automatically channeling spare change into micro-investment portfolios via an asynchronous event-driven model.

1.3 Target Audience and Market Segmentation

EuroGlobe Bank delivers tailored banking services across three primary demographic segments:

- **Individuals:** Ranging from local Lithuanian citizens seeking primary checking and mortgage solutions, to international residents and temporary stay groups (such as Erasmus students) requiring seamless identity verification and multi-currency support.
- **Businesses:** Small and Medium-sized Enterprises (SMEs) requiring robust treasury management, business card issuance, and multi-user transactional authorization workflows.
- **Groups and Communities:** Co-habitants, travel groups, and joint organizers who require collective funds management with built-in multi-signature approval rules.

1.4 System Complexity and Architectural Scope

EuroGlobe Bank is explicitly engineered beyond the scope of a standard CRUD (Create, Read, Update, Delete) application. The underlying architecture integrates sophisticated software engineering paradigms to handle complex financial operations safely:

- **Finite State Machines (FSM):** Orchestrate the complex, multi-stage lifecycles of mortgage applications, car leasing agreements, and card issuance states.
- **Event-Driven Architecture (EDA):** Drives real-time micro-investment calculations and notifications asynchronously upon the settlement of retail transactions, ensuring minimal latency on primary payment processing threads.
- **Consensus & Multi-Signature Engine:** Enforces multi-party consent protocols for vault transfers, managing pending transaction queues, expiration windows, and cryptographically verified approvals.

2. Business Requirements (MoSCoW + User Stories)

2.1 MoSCoW Prioritization Framework

Must Have (Critical Requirements)

1. **Multi-Track Account Registration:** Tailored onboarding flows supporting identity verification for local citizens, international residents (e.g., Erasmus students), and commercial entities (SMEs) with automated KYC/AML checks.
2. **Multi-Currency Basic Accounts and Instant Transfers:** Core ledger functionality enabling account management across multiple currencies alongside SEPA Instant processing capabilities.
3. **Smart Mortgages Engine:** Mortgage application, approval, and management lifecycle supporting dynamic fixed and variable interest calculations, alongside flexible early repayment schedules.
4. **Integrated Car Leasing:** End-to-end vehicle leasing module providing dynamic term customization, down-payment adjustments, and automated monthly billing.
5. **Multi-Signature Shared Vaults:** Group account structures requiring configured approval thresholds (e.g., 2 out of 3 designated members) prior to fund disbursement.
6. **Automated Micro-Investments:** Asynchronous transaction parsing that rounds up card purchase amounts to the nearest unit and routes residual balance into designated investment assets.
7. **Card Issuance and Management:** Lifecycle administration for both physical and virtual debit/credit cards, including PIN management, spending limits, and temporary freeze capabilities.

Should Have (Important Requirements)

1. **Investment Analytics Dashboard:** Graphical performance reporting, historical trends, asset allocation metrics, and yield estimation tools for micro-investment portfolios.

Could Have (Desirable Requirements)

1. **Manual Capital Investments:** Self-directed investment module allowing users to manually purchase fractional stocks, ETFs, or designated cryptocurrency pairs.
2. **Supplemental Insurance Packages:** Optional point-of-sale add-ons for extended vehicle warranties, property protection, and international travel coverage.

Won't Have (Out of Scope for Initial Release)

1. Unsecured Personal Consumer Loans.
2. Enterprise Payroll Processing Engine.
3. High-Frequency / Advanced Trading Terminal Features.

2.2 Detailed User Stories and Acceptance Criteria

Story 1: Smart Mortgage Application

- **User Story:** *As a prospective homeowner*, I want to compare and select between fixed and variable interest rates, *so that* I can choose a financing option tailored to my long-term financial risk tolerance.
- **Acceptance Criteria:**
    - The system must display a dynamic amortization schedule comparing estimated monthly payments between fixed and variable rates based on current central bank benchmarks.
    - The applicant must upload verified proof of income documentation and explicitly consent to automated credit agency background checks.
    - Upon submission, the mortgage application must transition into a *PENDING_REVIEW* state governed by the mortgage state machine.

Story 2: Shared Vault Withdrawal Approvals

- **User Story:** As a shared vault administrator, I want to enforce a multi-signature rule requiring 2 out of 3 co-signers to authorize withdrawals, so that group funds cannot be spent unilaterally.
- **Acceptance Criteria:**
    a. A shared vault must require a minimum of 3 linked members during creation to enable multi-signature policies.
    b. When a member initiates a withdrawal request, the vault state records a *PENDING_APPROVAL* status and notifies all designated signers immediately.
    c. If the second required signature is not registered within a strict 24-hour expiration window, the pending withdrawal request automatically cancels and funds remain locked.

3. Non-Functional Requirements (Measurable)

3.1 Performance Requirements

- **Transaction Latency:** The system must process standard account queries and payment transactions with a 95th percentile (p95) response time under 200 ms under normal operational loads.
- **Throughput:** Core ledger service must handle up to 500 financial transactions per second (TPS) without operational degradation.

3.2 Scalability Requirements

- **Concurrent Users:** The infrastructure must support at least 10,000 active concurrent user sessions across mobile and web channels without impacting system responsiveness.
- **Horizontal Elasticity:** Microservices powering transaction processing must scale horizontally in response to CPU/memory utilization spikes exceeding 70%.

3.3 Availability and Reliability

- **Service Uptime:** The system must maintain a minimum operational uptime of 99.9% (excluding scheduled maintenance windows communicated in advance).
- **Fault Isolation:** Non-critical components (such as investment analytics) must fail gracefully without impacting core payment processing or user account balances.

3.4 Security and Compliance Requirements

Data Encryption Standards

- **Encryption at Rest:** All confidential user data, transaction ledgers, and database backups must be encrypted using AES-256 standard encryption keys managed via dedicated hardware security modules (HSM).
- **Encryption in Transit:** All network traffic between client applications and backend systems must strictly enforce Transport Layer Security version 1.3 (TLS 1.3).

Authentication and Access Control

- **Two-Factor Authentication (2FA):** Mandatory 2FA enforcement using time-based one-time passwords (TOTP) or hardware keys for high-risk operations including fund transfers and credential alterations.
- **Strong Customer Authentication (SCA):** Full compliance with PSD2 SCA regulations for online card transactions and sensitive account data access.

Auditability and Compliance

- **Immutable Audit Logs:** All financial operations, administrative overrides, and system state transitions must be committed to an append-only, tamper-proof audit log.

- **Regulatory Frameworks:** Full adherence to General Data Protection Regulation (GDPR) guidelines for data privacy and local Know Your Customer (KYC) / Anti-Money Laundering (AML) regulatory compliance.

3.5 Disaster Recovery and Business Continuity

- **Recovery Point Objective (RPO):** Maximum allowable data loss window is bounded at 15 minutes through real-time database replication and transactional logging.
- **Recovery Time Objective (RTO):** Maximum allowable service interruption duration following a critical system disaster is capped at 4 hours.

4. Business Flows (Sequence Diagrams)

4.1 Smart Mortgage

The feasibility calculation requires a maximum Debt-to-Income (DTI) ratio of 35% and a maximum Loan-To-Value (LTV) limit of 85%, enforcing a minimum down payment of 15%.

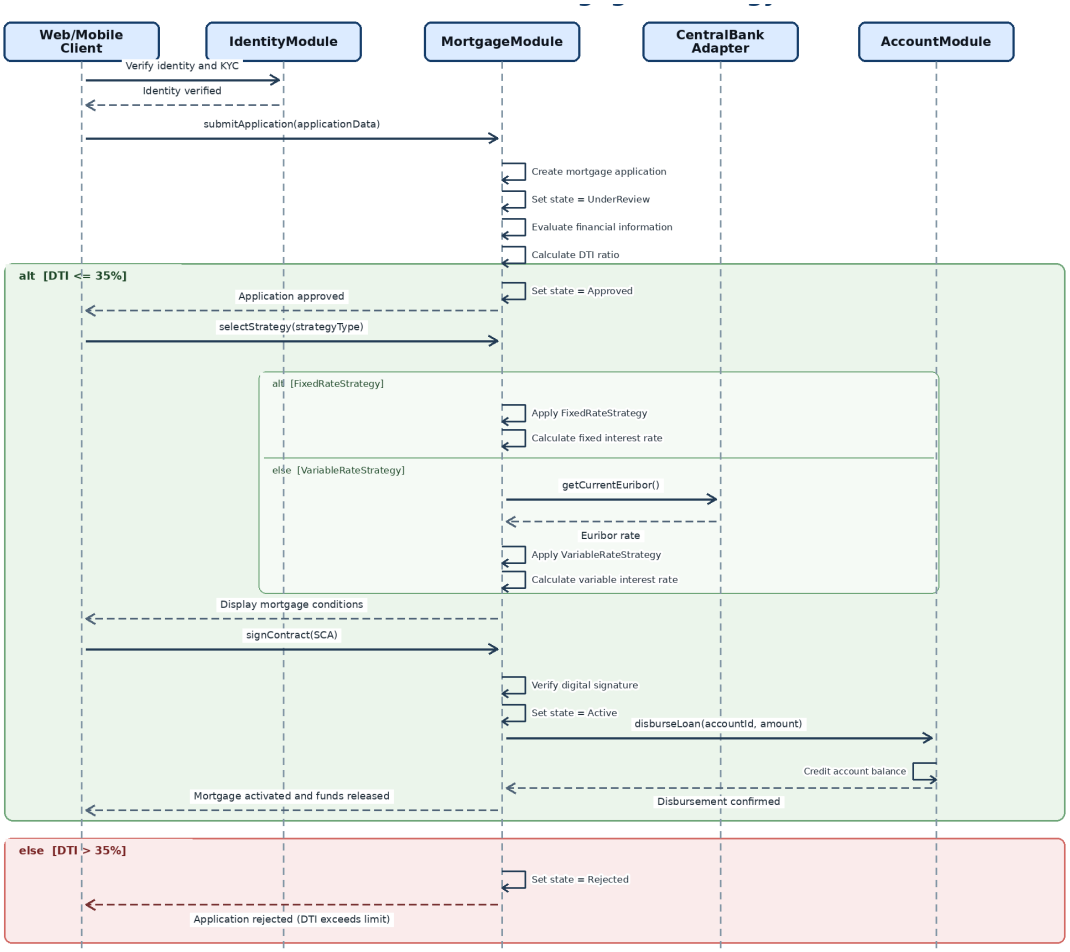


Figure 1: Smart Mortgage Business Flow

4.2 Car Leasing

The facade-orchestrated approval process blocks contract activation if any of the three synchronous services (Balance, Insurance, Dealer) returns a negative validation.

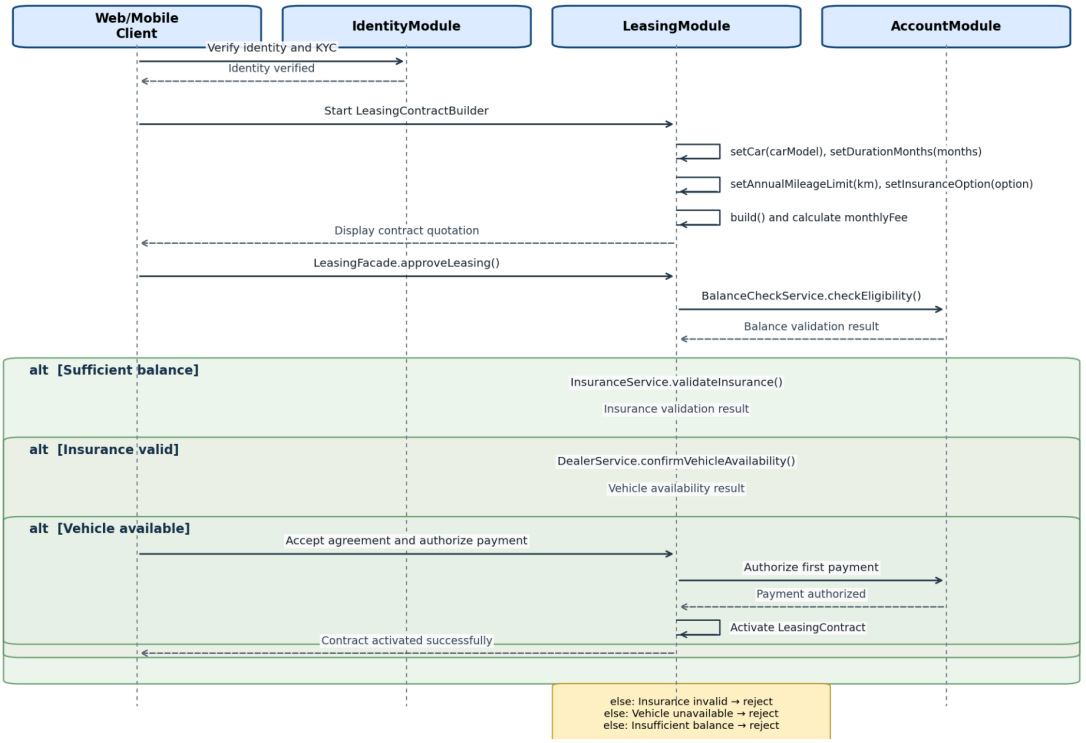


Figure 2: Car Leasing Approval Orchestration

4.3 Shared Vaults (Multi-signature)

A strict quorum of 2 signatures out of a total of 3 members (66.7%) is required. Any withdrawal request that fails to reach this quorum within an exact 24-hour time window expires automatically.

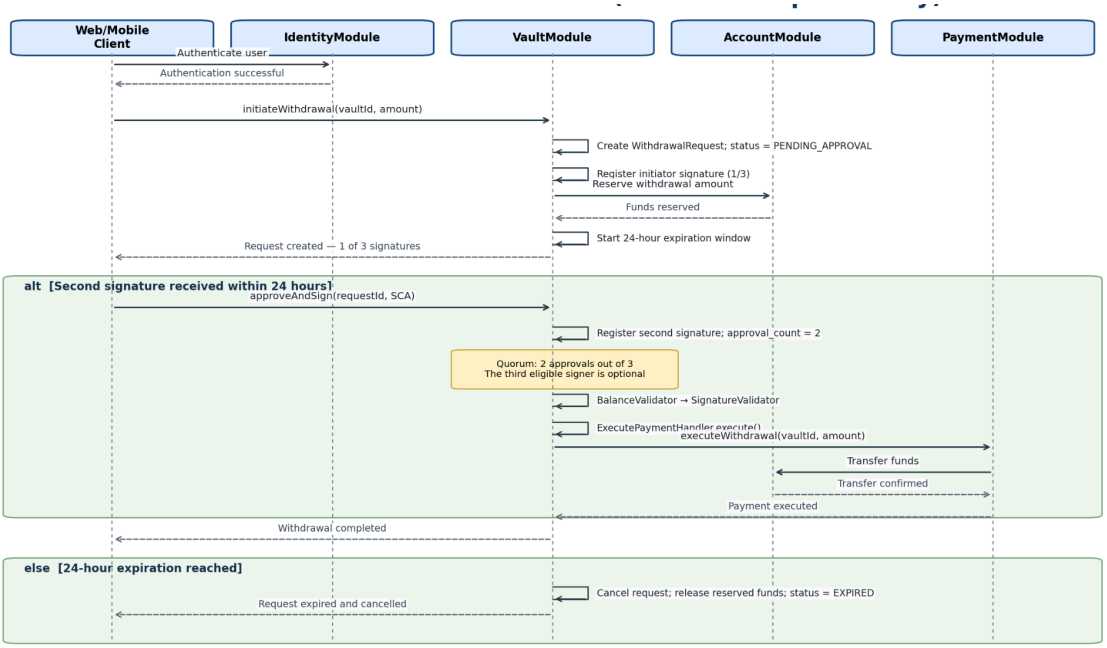


Figure 3: Shared Vault Multi-Signature Consensus Flow

4.4 Micro-investments

The round-up calculation applies a mathematical ceiling to the next whole Euro (ceil(baseAmount) - baseAmount). The execution is delegated to a background process (asynchronous) to prevent blocking the immediate response of the point-of-sale transaction.

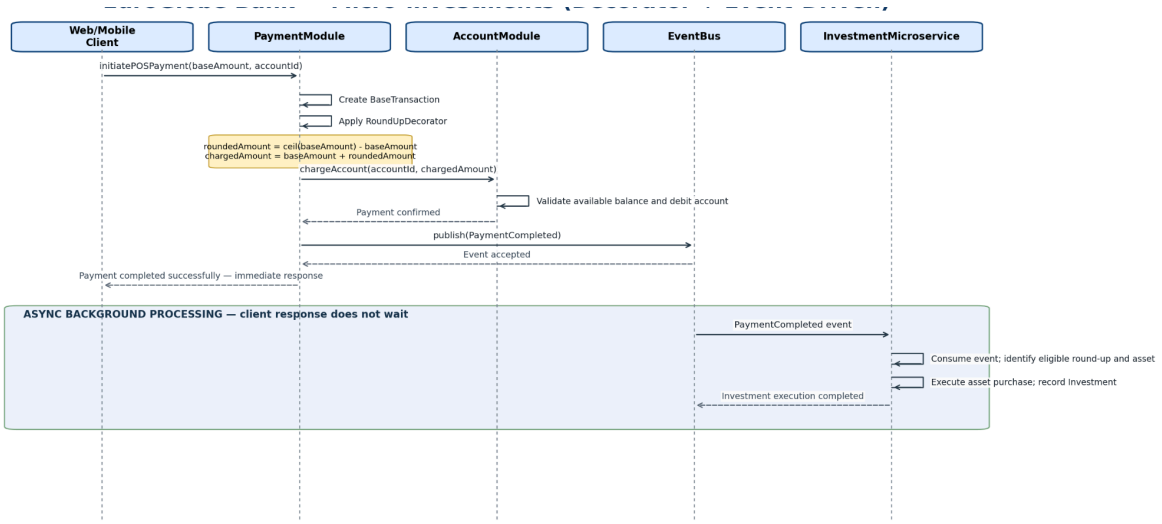


Figure 4: Event-Driven Micro-Investment Execution

5. High-Level Architecture.

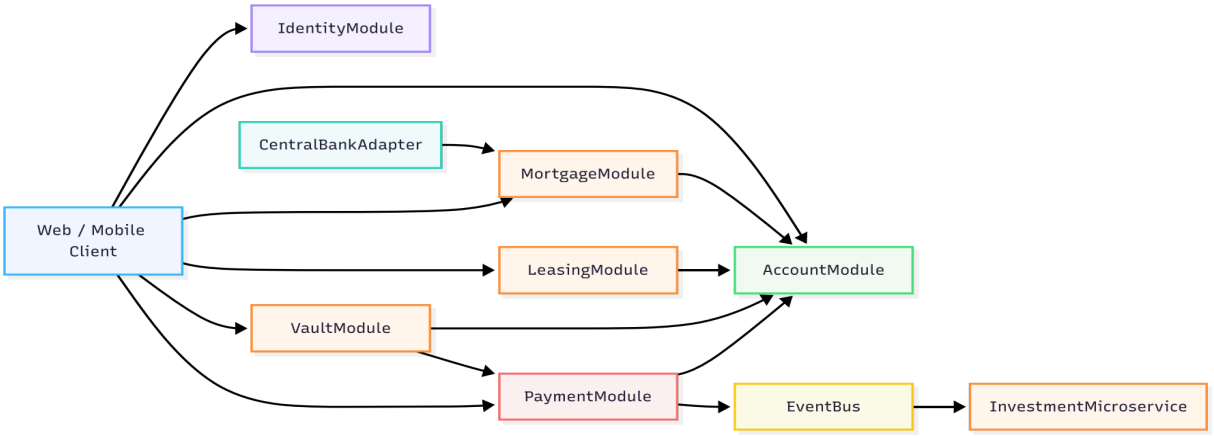


Figure 5. High-level architecture of the digital banking system.

5.1 Module Responsibilities

The Web / Mobile Client provides the user interface and sends requests to the system modules. The IdentityModule manages user and company registration, authentication, KYC checks, and card issuance. The AccountModule manages individual and group accounts, balances, and transfers. The MortgageModule assesses mortgage applications, calculates repayments, and manages mortgage states; it retrieves Euribor data through the CentralBankAdapter. The LeasingModule creates and approves car leasing contracts. The VaultModule manages shared vaults, approval signatures, and withdrawal requests. The PaymentModule validates and processes payments, then publishes payment events to the EventBus. The InvestmentMicroservice listens for these events and processes eligible round-up investments asynchronously.

6. Design Patterns

6.1 Strategy Pattern.

The Strategy pattern allows the MortgageModule to calculate mortgage interest using interchangeable algorithms. The FixedRateStrategy calculates interest using a fixed rate, while the VariableRateStrategy uses a variable rate based on the Euribor. The MortgageModule uses the selected strategy without needing to change its core logic.

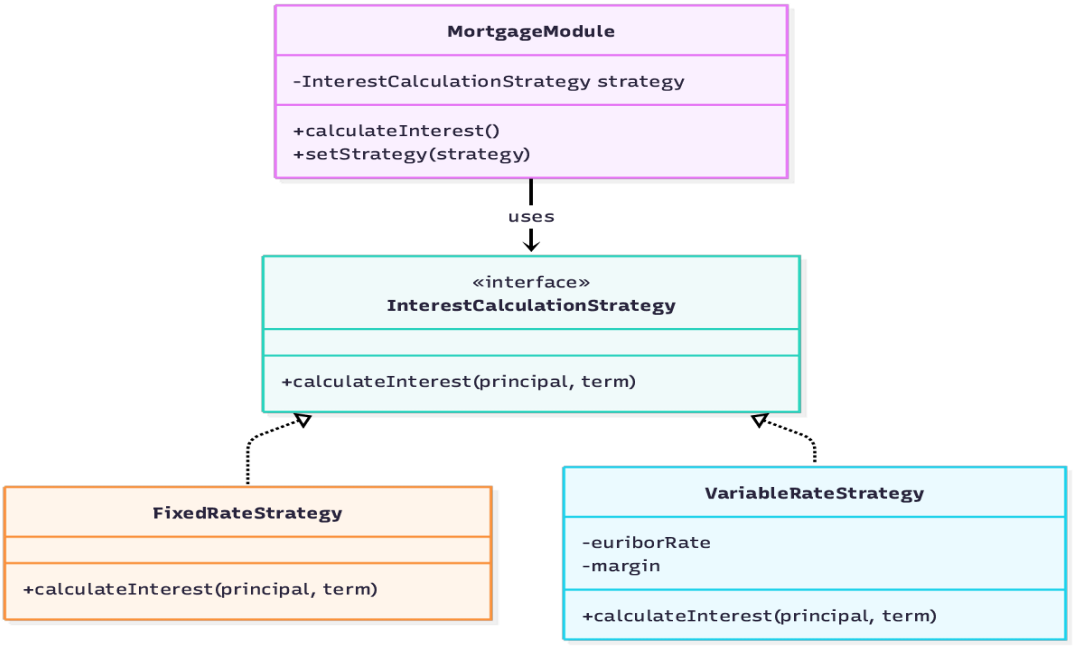


Figure 6. Strategy pattern for fixed and variable mortgage interest calculations.

6.2. Observer Pattern.

The Observer pattern notifies active variable-rate mortgages when the Euribor rate changes. The CentralBankAdapter provides the updated rate to the MortgageModule, which notifies each subscribed VariableMortgageObserver. A mortgage is unsubscribed when it is finalised.

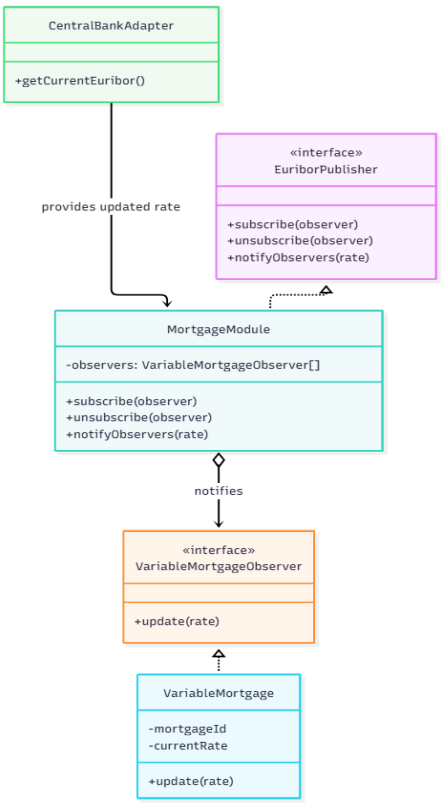


Figure 7. Observer pattern for Euribor updates to variable-rate mortgages.

6.3. State Pattern

The State pattern models the mortgage lifecycle and controls which actions are allowed in each state. A mortgage moves from UnderReview to Approved, then to Active. If payments are missed, it may enter Delinquent; it can later return to Active after regularisation or move to Closed when fully repaid or terminated.

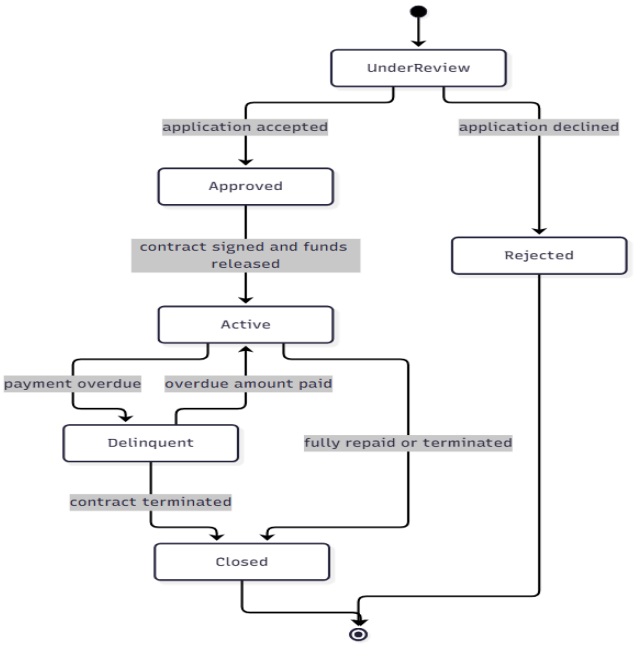


Figure 8. State diagram for the mortgage lifecycle.

6.4 Builder Pattern.

The Builder pattern creates a LeasingContract step by step and keeps contract construction separate from the rest of the LeasingModule. The LeasingContractBuilder configures the required details, such as the selected car, contract duration, annual mileage limit, and insurance option. The build() operation validates the configuration and returns a LeasingContract only when all required values are valid.

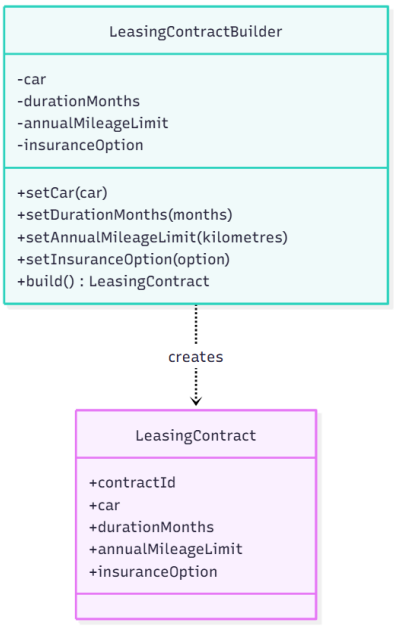


Figure 9. Builder pattern for creating leasing contracts.

6.5. Facade Pattern.

The Facade pattern provides a single entry point for approving a leasing application. The LeasingFacade coordinates the required checks and services, such as balance validation, insurance selection, and dealer confirmation, while hiding their internal complexity from the client.

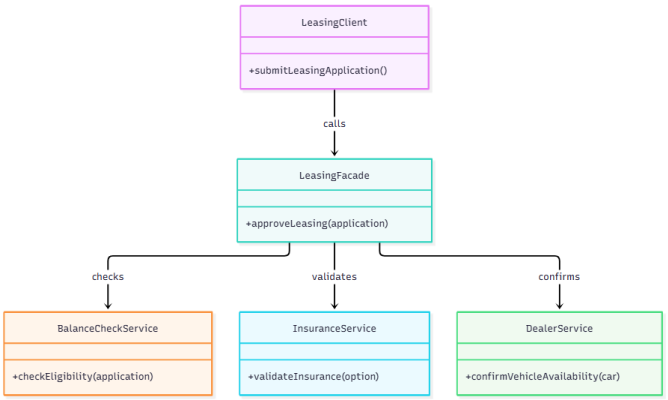


Figure 10. Facade pattern for approving a leasing application.

6.6. Composite Pattern.

The Composite pattern allows individual and group accounts to be handled through the same Account interface. An IndividualAccount represents a single account, while a GroupAccount contains member accounts. Both provide operations such as calculating a balance, so clients can work with either account type uniformly.

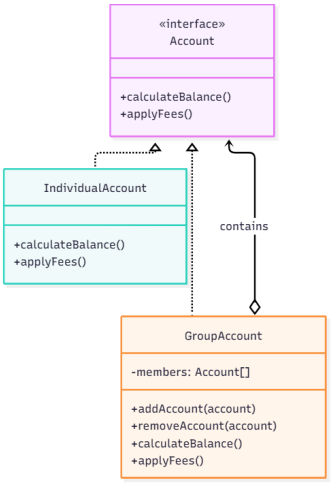


Figure 11. Composite pattern for individual and group accounts.

6.7. Chain of Responsibility Pattern.

The Chain of Responsibility pattern processes a shared vault withdrawal request through a sequence of handlers. The BalanceValidator checks that sufficient funds are available, the SignatureValidator verifies that the required member approvals have been collected, and the ExecutePaymentHandler processes the withdrawal. If a validation fails, the chain stops and the withdrawal is rejected.

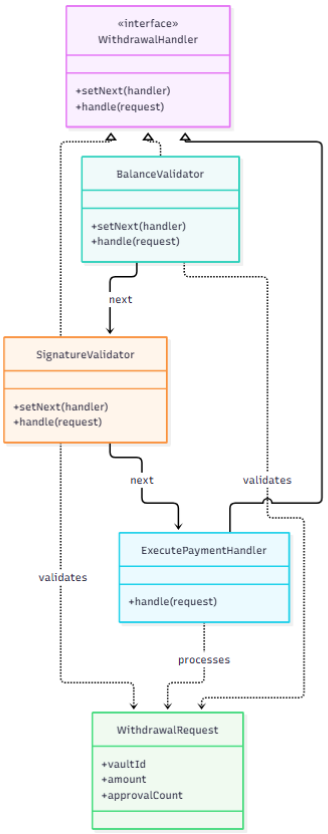


Figure 12. Chain of Responsibility for shared vault withdrawals.

6.8. Decorator Pattern

The Decorator pattern adds round-up investment behaviour to a transaction without changing the base transaction class. The RoundUpDecorator wraps a BaseTransaction, calculates the difference between the purchase amount and the next whole monetary unit, and makes that amount available for the investment process.

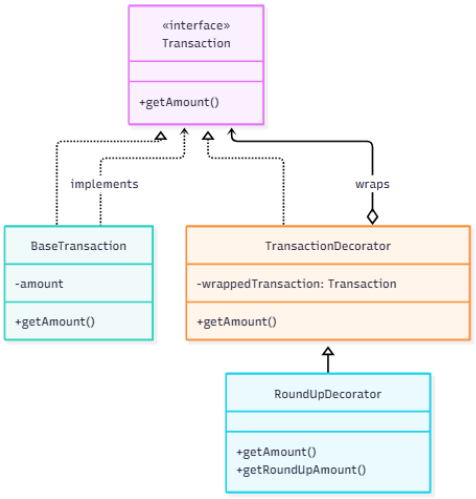


Figure 13. Decorator pattern for transaction round-up calculations.

6.9 Event-Driven Architecture Pattern

Event-driven architecture decouples payment processing from automatic micro-investments. After a payment is completed, the PaymentModule publishes a PaymentCompleted event to the EventBus. The InvestmentMicroservice consumes the event and processes any eligible round-up investment asynchronously, so investment processing does not delay the payment response.



Figure 14. Event-driven processing of round-up investments after payment completion.

7. Data Model (Tables with Fields + Types + Relationships)

The following diagram Figure 15 presents the core relational data model for EuroGlobe Bank. At the core of the system is the Account entity, acting as the central ledger. It is linked to User profiles (which manage KYC compliance and legal types) and serves as the foundation for all financial products:

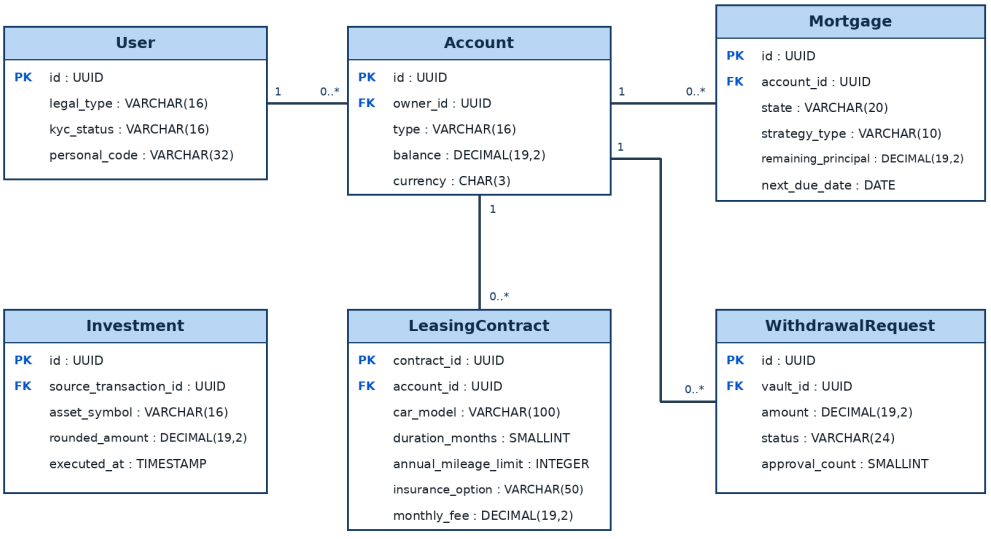


Figure 15. EuroGlobe Bank Core Relational Data Model Schema.

8. UI/UX LoFi Wireframes (All Screens + Transitions)

This section specifies the complete low-fidelity (LoFi) frontend design and transition logic for EuroGlobe Bank. All interfaces are systematically mapped to domain entities, finite state machines, and architectural design patterns. Screens are displayed in paired layouts for streamlined comparative navigation.

8.0 Global Navigation Architecture & Screen Flow

The client interface orchestrates six core functional areas: Multi-Track Onboarding, Unified Dashboard, Smart Mortgages, Vehicle Leasing, Multi-Signature Shared Vaults, and Cards with Round-Up Micro-Investments. Figure 16 presents the global transition topology.

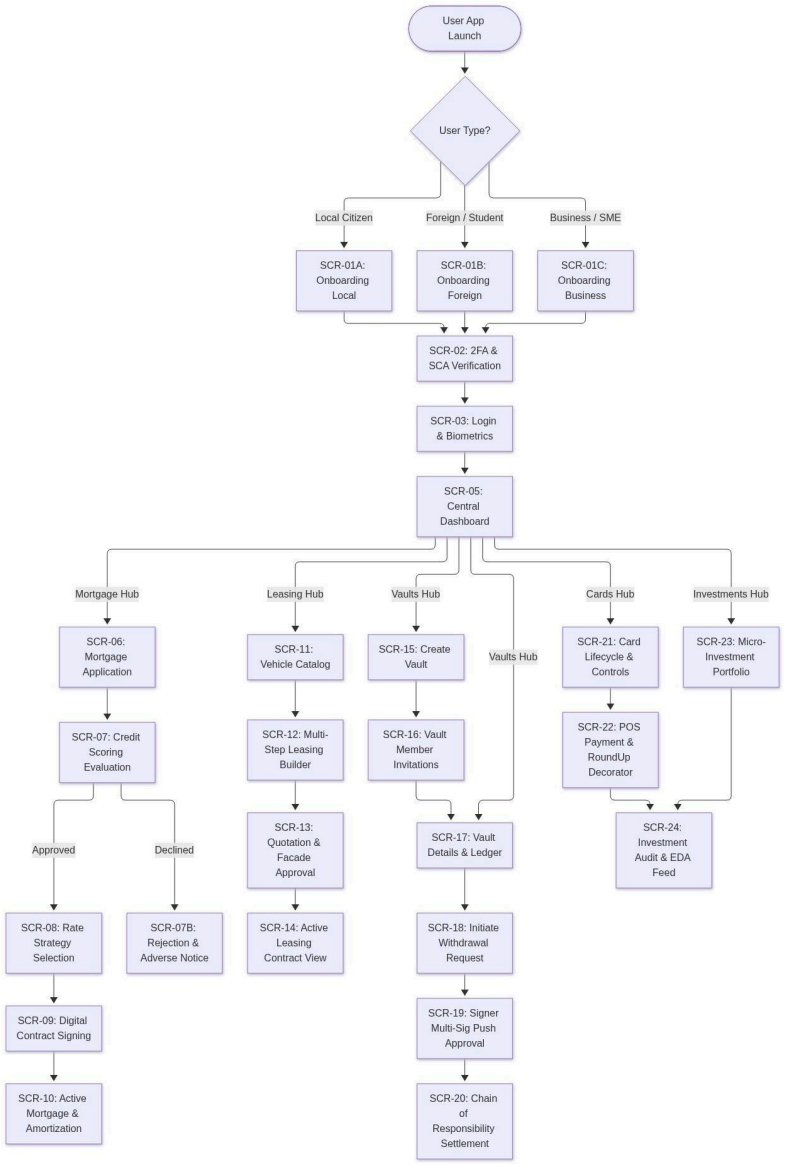


Figure 16. Global Screen Navigation Architecture and State-Driven Flow of EuroGlobe Bank.

8.1 Module 1: Onboarding & Multi-Track Registration

Supports specialized KYC tracks for local citizens, international students, and commercial entities (SMEs) with mandatory PSD2 2FA/SCA.

SCR-01A: Local Citizen Onboarding

```
+-----------------------------------------+
| EuroGlobe | Local Citizen      [Step 1/3]|
+-----------------------------------------+
| Full Name: [ Jonas Kazlauskas          ] |
| Personal Code: [ 39405120000           ] |
| DoB: [1994-05-12]  Gender: [Male v     ] |
| Address: [ Gedimino pr. 12-4, Vilnius  ] |
| Email: [ jonas.k@euroglobe.lt          ] |
| Phone: [ +370 60012345                 ] |
+-----------------------------------------+
| Doc: [ Upload ID / Passport ] (JPG/PDF)  |
| Status: [ id_card_front.jpg - OK ]       |
| [X] Consent KYC/AML & Terms of Service   |
+-----------------------------------------+
| [ Back ]               [ Next: Set 2FA ] |
+-----------------------------------------+
```

SCR-01B: Foreign Student Onboarding

```
+-----------------------------------------+
| EuroGlobe | Foreign Resident   [Step 1/3]|
+-----------------------------------------+
| Full Name: [ Carlos Ruiz Garcia        ] |
| Nationality: [ Spanish (ES)          v ] |
| Passport: [ PA9823411                  ] |
| Status: (*) Erasmus Student  ( )Resident |
| University: [ Vilnius University (MIF) ] |
| LT Address: [ Sauleteckio al. 9, VNO   ] |
| Tax Country: [ Spain                   ] |
+-----------------------------------------+
| Docs: - Passport Scan: [ passport.pdf OK]|
|       - Erasmus Proof: [ learning.pdf OK]|
+-----------------------------------------+
| [ Back ]               [ Next: Set 2FA ] |
+-----------------------------------------+
```

SCR-01C: Commercial (SME) Onboarding

```
+-----------------------------------------+
| EuroGlobe | Business Account   [Step 1/3]|
+-----------------------------------------+
| Company: [ Baltic FinTech Solutions UAB] |
| Reg Code: [ 305891234                  ] |
| VAT Number: [ LT100014234511           ] |
| Address: [ Konstitucijos pr. 21, VNO   ] |
| Industry: [ 62.01 - Programming      v ] |
+-----------------------------------------+
| Legal Representative:                    |
| Name: [ Viktoras Navickas              ] |
| Personal ID: [ 38801020000             ] |
| Certificate: [ registry_extract.pdf OK ] |
+-----------------------------------------+
| [ Back ]             [ Next: Corp 2FA ] |
+-----------------------------------------+
```

SCR-02: Strong Customer Auth (2FA)

```
+-----------------------------------------+
| EuroGlobe | Security: 2FA/SCA  [Step 2/3]|
+-----------------------------------------+
| Scan QR Code with Authenticator App:     |
|          +-------------------+           |
|          |  [ QR CODE IMAGE ]|           |
|          |    TOTP Secret    |           |
|          +-------------------+           |
| Setup Key: [ KZX4-9912-EGBK-0021 ]       |
|                                          |
| 6-Digit Code: [2] [8] [4] [1] [9] [0]    |
| [X] Enable Biometrics (FaceID / TouchID) |
+-----------------------------------------+
| [ Verify & Finish Onboarding ]           |
+-----------------------------------------+
```

Table 8.1: UI Elements & Transition Specification — Onboarding Module

Screen ID	Element / Control	Data Model Binding	Action / Logic	Target Screen
SCR-01A	Button: [Next: Set 2FA]	User(legal_type='local', kyc_status='PENDING')	Validates LT ID checksum and initiates automated AML check.	SCR-02
SCR-01B	Button: [Next: Set 2FA]	User(legal_type='foreign', kyc_status='PENDING')	Verifies international passport and academic certificate.	SCR-02

SCR-01C	Button: [Next: Corp 2FA]	User(legal_type='business', kyc_status='PENDING')	Queries State Registers database for legal representative.	SCR-02
SCR-02	Button: [Verify & Finish]	User.two_factor_secret, kyc_status='VERIFIED'	Validates RFC 6238 TOTP; initializes primary Account.	SCR-05

8.2 Module 2: Unified Central Dashboard (Composite Pattern)

Aggregates individual and group balances via Composite Account interface alongside live product cards.

SCR-05: Unified Central Multi-Product Dashboard

```
+-------------------------------------------------------------------+
| EuroGlobe Bank       [Notification: 2] [EUR/USD: 1.08]  (User: Jonas)      |
+-------------------------------------------------------------------+
| COMPOSITE LIQUID BALANCE: EUR 14,850.40                           |
|  |- Checking (IndividualAccount #LT02..101):         EUR  8,420.40 |
|  |- Flat Vilnius Vault (GroupAccount #LT99..402):     EUR  1,230.00 |
|  |- SME Treasury (GroupAccount #LT44..901):           EUR  5,200.00 |
+-------------------------------------------------------------------+
| ACTIVE PRODUCTS AT A GLANCE:                                      |
| [*] SMART MORTGAGE (#MTG-8821)                          [ ACTIVE   ] |
|     Principal: EUR 185,420.00 | Rate: 4.15% (Var) | Next: EUR 942.10|
|     [ View Details / Amortize Early ]                             |
| [*] CAR LEASING (#LSG-4019)                             [ ACTIVE   ] |
|     Audi Q4 e-tron | Monthly: EUR 429.00 | Next Due: 2026-11-28     |
|     [ Manage Leasing Contract ]                                   |
| [*] MICRO-INVESTMENT PORTFOLIO                          [ EDA Feed ]  |
|     Total Saved via Round-Ups: EUR 1,420.80 (+14.2% yield)        |
|     Allocation: BTC 80% | S&P 500 ETF 20%                         |
| [*] ACTIVE CARDS                                                  |
|     Debit (..4092) - ACTIVE | Daily Limit: EUR 1,500.00           |
|     [ + New Transfer ]   [ Shared Vaults ]   [ Freeze Card ]      |
+-------------------------------------------------------------------+
```

Table 8.2: UI Elements & Transition Specification — Central Dashboard

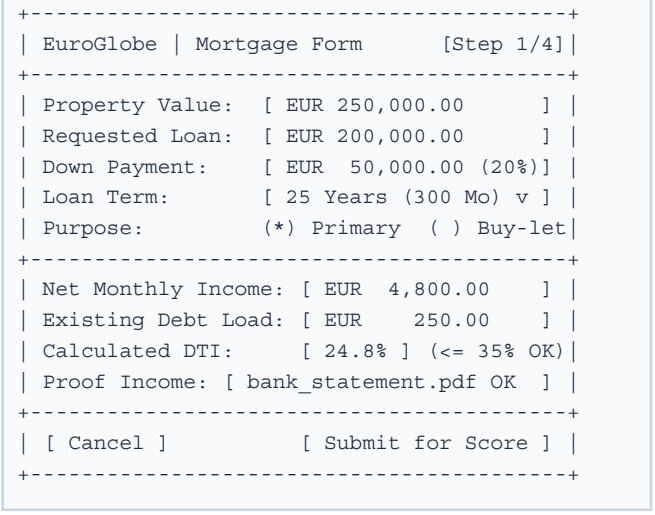
Screen ID	Element / Control	Data Model Binding	Action / Logic	Target Screen
SCR-05	Button: [View Details]	Mortgage.id, Mortgage.state	Queries MortgageModule for amortization and status.	SCR-10

SCR-05	Button: [Manage Leasing]	LeasingContract.id	Loads active lease terms and insurance coverage.	SCR-14
SCR-05	Button: [Shared Vaults]	GroupAccount.id, VaultMember	Navigates to multi-signature vault overview.	SCR-17
SCR-05	Button: [Freeze Card]	Card.status	Toggles card between ACTIVE and FROZEN instantly.	SCR-21

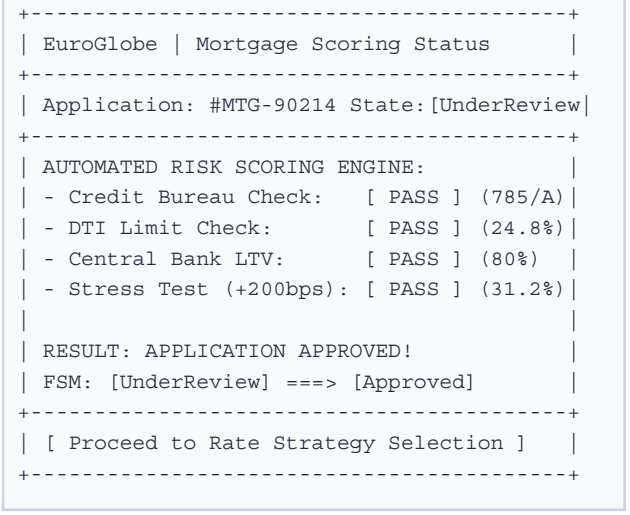
8.3 Module 3: Smart Mortgage Engine (State, Strategy & Observer Patterns)

Implements full FSM lifecycle (UnderReview -> Approved -> Active <-> Delinquent -> Closed) with Strategy and Observer patterns.

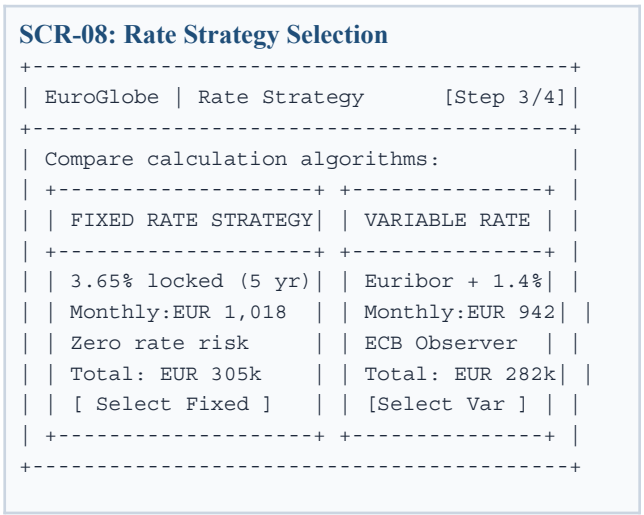
SCR-06: Mortgage Application Form



SCR-07: Credit Scoring & Approval



SCR-08: Rate Strategy Selection



SCR-09: Digital Contract Signing



```
| Note: VariableRateStrategy recalculates  |
| automatically on Euribor updates (Obs).  |
+------------------------------------------+
```

```
| [ Sign Contract & Disburse Principal ]   |
+------------------------------------------+
```

SCR-10: Active Mortgage Management & Early Amortization

```
+------------------------------------------+
| EuroGlobe | Active Mortgage Lifecycle    |
+------------------------------------------+
| Loan: #MTG-90214           State: [ACTIVE]|
| Principal: EUR 198,420 | Quota: EUR 942.1|
| Rate: 4.15% (Var)      | Due: 2026-12-01 |
+------------------------------------------+
| EARLY REPAYMENT / AMORTIZATION:          |
| Amount to Repay: [ EUR 10,000.00       ] |
| Strategy Effect:                         |
|  (*) Option 1: Reduce Amortization Term  |
|      New Term: 274 Mo (-22 mo / -EUR 8k) |
|  ( ) Option 2: Reduce Monthly Quota      |
|      New Quota: EUR 894.40 (-EUR 47.7/mo)|
+------------------------------------------+
| [ Confirm Early Repayment (Amortize) ]   |
+------------------------------------------+
```

Table 8.3: UI Elements & Transition Specification — Smart Mortgage Module

Screen ID	Element / Control	Data Model Binding	Action / Logic	Target Screen
SCR-06	Button: [Submit for Score]	Mortgage(state='UnderReview')	Runs scoring algorithms; transitions to UnderReview.	SCR-07
SCR-07	Button: [Proceed to Strategy]	Mortgage(state='Approved')	Advances state from UnderReview to Approved.	SCR-08
SCR-08	Button: [Select Var / Fixed]	Mortgage.strategy_type	Configures Strategy and registers Observer.	SCR-09
SCR-09	Button: [Sign & Disburse]	Mortgage(state='Active')	Signs contract; funds disbursed; transitions to Active.	SCR-10
SCR-10	Button: [Confirm Amortize]	Amortization(applied_to)	Applies early repayment; recalculates schedule.	SCR-10 (Updated)

8.4 Module 4: Integrated Vehicle Leasing (Builder & Facade Patterns)

Utilizes LeasingContractBuilder for step-by-step assembly and LeasingFacade for 3-service atomic validation.

SCR-11: Vehicle Catalog

```
+-----------------------------------------+
| EuroGlobe | Vehicle Leasing Catalog     |
+-----------------------------------------+
| Filter: [ All Makes v ] [ Electric v ]  |
+-----------------------------------------+
| [IMG] Audi Q4 45 e-tron (286 HP, EV)    |
|  Retail: EUR 56,200 | From EUR 429/mo   |
|  Stock: 3 units | Dealer: Audi Vilnius  |
|  [ Configure Leasing with Builder ]     |
|-----------------------------------------|
| [IMG] Tesla Model 3 Long Range (EV)     |
|  Retail: EUR 49,990 | From EUR 389/mo   |
|  Stock: 5 units | Dealer: Tesla Vilnius |
|  [ Configure Leasing with Builder ]     |
+-----------------------------------------+
```

SCR-12: Leasing Builder Wizard

```
+-----------------------------------------+
| EuroGlobe | Leasing Builder    [Step 2/3]|
+-----------------------------------------+
| Audi Q4 45 e-tron (VIN: WAUZZZGE8PB0123) |
|                                         |
| Duration: ( )24 Mo  (*)36 Mo  ( )48 Mo  |
| Annual Km: ( )15k km (*)20k km ( )30k km |
| Insurance: (*)Full Kasko  ( )Franchise  |
| Down Payment: [ EUR 5,620.00 (10%)    ] |
+-----------------------------------------+
| Real-Time Calculated: EUR 429.00 / month |
+-----------------------------------------+
| [ Cancel ]      [ Build & Review Quote ] |
+-----------------------------------------+
```

SCR-13: Facade Verification

```
+-----------------------------------------+
| EuroGlobe | Facade Approval Pipeline    |
+-----------------------------------------+
| Contract (#LSG-4019): Audi Q4 | 36 Mo    |
| Km: 20k/yr | Ins: Full Kasko | EUR 429/mo|
+-----------------------------------------+
| LEASING FACADE VERIFICATION PIPELINE:   |
| 1. BalanceCheckService: [ OK ] Funds OK |
| 2. InsuranceService:    [ OK ] ERGO Auth |
| 3. DealerService:       [ OK ] VIN Locked|
|                                         |
| Facade Result: ALL SERVICES APPROVED    |
+-----------------------------------------+
| [ Accept Agreement & Authorize Initial ] |
+-----------------------------------------+
```

SCR-14: Active Lease View

```
+-----------------------------------------+
| EuroGlobe | Active Lease (#LSG-4019)    |
+-----------------------------------------+
| Status: [ ACTIVE ]     Reg: LTH-401     |
| Monthly: EUR 429.00    Next: 2026-11-28 |
| Direct Debit: Checking Account #LT02..101|
+-----------------------------------------+
| Mileage Tracker:                        |
| [=====>-------------] 2,420 / 20,000 km |
+-----------------------------------------+
| Docs: [ Contract PDF ] [ Insurance Card ]|
| Actions: [ Report Incident / Claim ]    |
+-----------------------------------------+
```

Table 8.4: UI Elements & Transition Specification — Vehicle Leasing Module

Screen ID	Element / Control	Data Model Binding	Action / Logic	Target Screen
SCR-11	Button: [Configure Builder]	LeasingContract.car_model	Instantiates LeasingContractBuilder with selected car.	SCR-12

SCR-12	Button: [Build & Review]	LeasingContractBuilder.build()	Validates constraints; compiles contract object.	SCR-13
SCR-13	Button: [Accept Agreement]	LeasingFacade.approveLeasing( )	Orchestrates Balance, Insurance, and Dealer services.	SCR-14
SCR-14	Button: [Report Claim]	LeasingContract.id	Initiates accident claim handling flow.	External / Modal

8.5 Module 5: Multi-Signature Shared Vaults (Chain of Responsibility)

Enforces 2-of-3 signature quorum, 24-hour expiration, and BalanceValidator -> SignatureValidator -> ExecutePaymentHandler chain.

SCR-15: Vault Creation

```
+-----------------------------------------+
| EuroGlobe | Create Shared Vault         |
+-----------------------------------------+
| Vault Name:   [ Flat Vilnius Rent     ] |
| Currency:     [ EUR (Euro)          v ] |
| Initial Dep.: [ EUR  300.00           ] |
+-----------------------------------------+
| Multi-Sig Consensus Rules:              |
| - Minimum Members: 3 Co-signers         |
| - Required Quorum: 2 of 3 Signatures    |
| - Window: Strict 24-Hour Expiration     |
+-----------------------------------------+
| [ Next: Invite Linked Co-signers ]      |
+-----------------------------------------+
```

SCR-16: Member Invitations

```
+-----------------------------------------+
| EuroGlobe | Invite Members     [Step 2/2]|
+-----------------------------------------+
| Vault: "Flat Vilnius Rent" (Quorum: 2/3) |
|                                         |
| Member 1 (Admin): Jonas K.    [ACCEPTED] |
| Member 2: [ laura.m@euroglobe.lt      ] |
|           Status: [INVITATION PENDING]  |
| Member 3: [ tomas.v@euroglobe.lt      ] |
|           Status: [INVITATION PENDING]  |
+-----------------------------------------+
| [ Finalize Vault Creation ]             |
+-----------------------------------------+
```

SCR-17: Vault Ledger

```
+-----------------------------------------+
| EuroGlobe | Vault: Flat Vilnius Rent    |
+-----------------------------------------+
| Vault Account: #LT99..402 (GroupAccount) |
| COLLECTIVE BALANCE: EUR 1,850.00        |
| Policy: 2-of-3 Signatures Required      |
+-----------------------------------------+
| ACTIVE REQUESTS:                        |
| [!] REQ #WTH-5510: Rent (EUR 900.00)    |
|     Approvals: [ 1 / 2 Collected ]      |
|     Expires in: 14h 22m [ View Progress ]|
+-----------------------------------------+
| [ Initiate Withdrawal ] [ Manage Members]|
+-----------------------------------------+
```

SCR-18: Request Withdrawal

```
+-----------------------------------------+
| EuroGlobe | Request Fund Disbursement   |
+-----------------------------------------+
| Vault Balance: EUR 1,850.00 Available   |
| Amount:        [ EUR  900.00          ] |
| Recipient IBAN:[ LT12 7044 0001 2345 6789] |
| Beneficiary:   [ UAB Vilnius Real Est. ] |
| Reference:     [ Nov 2026 Rent Apt 4   ] |
+-----------------------------------------+
| Terms: Funds reserved immediately.      |
| Requires 1 additional co-signer in 24h. |
+-----------------------------------------+
| [ Cancel ]       [ Create & Sign (1/2) ] |
+-----------------------------------------+
```

SCR-19: Signer Mobile Approval

```
+-----------------------------------------+
| EuroGlobe Mobile | ACTION REQUIRED: SCA  |
+-----------------------------------------+
| Signature Request from: Jonas Kazlauskas |
| Vault: Flat Vilnius Rent                |
| Amount: EUR 900.00 | To: Vilnius Real Est|
| STATUS: [ 1 of 2 Signatures Collected ]  |
| Timer: [ 14 hours 22 minutes remaining ] |
+-----------------------------------------+
| [ REJECT REQUEST ]  [ APPROVE & SIGN SCA]|
+-----------------------------------------+
```

SCR-20: Chain Settlement Result

```
+-----------------------------------------+
| EuroGlobe | Settlement Engine Result     |
+-----------------------------------------+
| CHAIN OF RESPONSIBILITY EXECUTION:       |
| [1] BalanceValidator:    [ PASS ] EUR 1.8k|
| [2] SignatureValidator:  [ PASS ] 2/2 Sig|
| [3] ExecutePaymentHandler:[ DONE ] SEPA  |
+-----------------------------------------+
| Status: [ APPROVED & SETTLED ]           |
| Tx Hash: #0x9bf4102c                     |
| [ Return to Vault Dashboard ]            |
+-----------------------------------------+
```

Table 8.5: UI Elements & Transition Specification — Shared Vault Module

Screen ID	Element / Control	Data Model Binding	Action / Logic	Target Screen
SCR-15	Button: [Next: Invite]	Account(type='group')	Initializes group vault structure.	SCR-16
SCR-16	Button: [Finalize Vault]	VaultMember(role)	Enforces 3-member minimum rule.	SCR-17
SCR-17	Button: [Initiate Withdrawal]	WithdrawalRequest.vault_id	Opens disbursement modal.	SCR-18
SCR-18	Button: [Create & Sign (1/2)]	WithdrawalRequest(status='PENDING')	Reserves funds; sends signer push alerts.	SCR-17
SCR-19	Button: [APPROVE & SIGN]	WithdrawalApproval(decision='APPROVED')	Registers 2nd signature; triggers Chain.	SCR-20
SCR-19	Button: [REJECT]	WithdrawalApproval(decision='REJECTED')	Cancels request; releases reserved funds.	SCR-17

8.6 Module 6: Cards, Payments & Micro-Investments (Decorator & EDA)

Processes spare change via RoundUpDecorator; settles asynchronously via PaymentCompleted event.

SCR-21: Card Security & Round-Up

```
+-----------------------------------------+
| EuroGlobe | Card Management & Security  |
+-----------------------------------------+
| [CARD] DEBIT ..5592 | Jonas Kazlauskas  |
| Status: [ ACTIVE ]  | Limit: EUR
1,500/day|
+-----------------------------------------+
| Controls: [ Freeze Card (Instant Lock) ] |
|           [ Change Card PIN via SCA    ] |
+-----------------------------------------+
| ROUND-UP MICRO-INVESTMENTS:              |
| Status: [ ENABLED ] | Multiplier: [ 1x ] |
| Asset: (*) Bitcoin (BTC)  ( ) S&P 500 ETF|
+-----------------------------------------+
```

SCR-22: POS Receipt & RoundUpDecorator

```
+-----------------------------------------+
| EuroGlobe | Payment Receipt & Round-Up  |
+-----------------------------------------+
| Merchant: Caffeine Roasters Vilnius      |
| Base Amount (BaseTransaction):   EUR 2.40|
| Round-Up (RoundUpDecorator):    +EUR 0.60|
| TOTAL CHARGED TO CHECKING:       EUR 3.00|
+-----------------------------------------+
| EDA Event: PaymentCompleted {            |
|   tx_id: "TX-7782109", round_up: 0.60    |
| } ==> Published to EventBus in 18 ms!    |
+-----------------------------------------+
| [ View in Ledger ]   [ Go to Portfolio ] |
+-----------------------------------------+
```

SCR-23: Micro-Investment Portfolio

```
+-----------------------------------------+
| EuroGlobe | Micro-Investment Portfolio  |
+-----------------------------------------+
| TOTAL PORTFOLIO: EUR 1,420.80 (+14.2%)   |
| Cumulative Round-Ups: EUR 1,244.40       |
+-----------------------------------------+
| ALLOCATION:                              |
| 1. Bitcoin (BTC): EUR 1,136.64 (+16.8%)  |
| 2. S&P 500 ETF:   EUR   284.16 (+5.4%)   |
+-----------------------------------------+
| [ Adjust Allocation ] [ Liquidate Assets]|
+-----------------------------------------+
```

SCR-24: EDA Event Ledger

```
+-----------------------------------------+
| EuroGlobe | Event-Driven Investment Feed |
+-----------------------------------------+
| 2026-11-05 | Caffeine VNO (Tx #TX-7782109|
|  Round-Up: EUR 0.60 -> 0.0000075 BTC OK  |
| 2026-11-04 | Maxima Market (Tx #TX-778104|
|  Round-Up: EUR 0.75 -> 0.0000094 BTC OK  |
+-----------------------------------------+
| Status: ALL ASYNC ORDERS EXECUTED        |
| [ Export Tax CSV Statement ]             |
+-----------------------------------------+
```

Table 8.6: UI Elements & Transition Specification — Cards & Investments Module

Screen ID	Element / Control	Data Model Binding	Action / Logic	Target Screen
SCR-21	Button: [Freeze Card]	Card.status	Toggles card between ACTIVE and FROZEN.	SCR-21 (Updated)
SCR-22	Button: [Go to Portfolio]	Investment.source_transaction_id	Navigates to micro-investment holdings.	SCR-23
SCR-23	Button: [Adjust Allocation]	Investment(asset_symbol)	Modifies round-up asset weighting.	SCR-23 (Modal)

SCR-24	Button: [Export CSV]	Investment.executed_at	Downloads compliant tax transaction report.	File Download

9. API Contract

Base path: /api. All requests and responses use JSON. Authentication uses a simple session token returned by /auth/login. Amounts are expressed in euros with two decimals (e.g., 12.50). Currency is EUR by default.

9.1 Errors

Standard HTTP status codes are used. No custom error codes are needed.

HTTP	When
400	Invalid request data (missing or wrong fields)
401	Not logged in or invalid token
403	Not allowed (e.g., not a vault member)
404	Resource not found
409	Duplicate or invalid state (e.g., already approved)
422	Business rule failed (insufficient funds, KYC pending)
500	Server error

9.2 Endpoints

12 endpoints cover all core features.

Method	Endpoint	Purpose	Auth
POST	/api/users/register	Register user (local/foreign/business) with basic KYC	No
POST	/api/auth/login	Login and get session token	No
POST	/api/accounts	Create individual or group account	Yes
GET	/api/accounts/{id}	Get account balance and details	Yes
POST	/api/accounts/{id}/transfers	Make a transfer to another IBAN	Yes
POST	/api/mortgages	Create mortgage application	Yes
POST	/api/mortgages/{id}/rate-decision	Choose fixed or variable rate	Yes
POST	/api/mortgages/{id}/amortize	Make an early repayment	Yes
POST	/api/leasing	Create leasing contract	Yes
POST	/api/leasing/{id}/approve	Approve leasing contract	Yes
POST	/api/vaults	Create shared vault	Yes
POST	/api/vaults/{id}/withdrawals	Create withdrawal request	Yes
POST	/api/vaults/{id}/withdrawals/{wid}/approve	Approve or reject withdrawal	Yes
POST	/api/cards	Issue a card	Yes

9.3 Request/Response examples

Register:

POST /api/users/register
{ "legal_type": "foreign", "full_name": "Ana", "email": "ana@x.com",
"password": "secret123", "country": "LT",
"kyc": { "document_type": "PASSPORT", "document_number": "X123",
"consent": true } }
Response 201: { "user_id": "usr_1", "status": "pending_kyc" }

Login:

POST /api/auth/login
{ "email": "ana@x.com", "password": "secret123" }
Response 200: { "token": "abc123", "user_id": "usr_1" }

Create account:

POST /api/accounts
{ "type": "individual" | "group", "currency": "EUR", "members": [] }
Response 201: { "account_id": "acc_1", "balance": 0.00, "status": "active" }

Transfer:

POST /api/accounts/{id}/transfers
{ "to_iban": "LT12...", "amount": 25.50, "currency": "EUR" }
Response 200: { "transfer_id": "trf_1", "status": "completed" }
Errors: 400 invalid data, 403 not owner, 422 insufficient funds

Mortgage:

POST /api/mortgages
{ "property_value": 250000, "down_payment": 50000, "amount": 200000,
"term_months": 300, "rate_type": "fixed|variable" }
Response 201: { "mortgage_id": "mtg_1", "status": "pending_review" }

POST /api/mortgages/{id}/rate-decision
{ "rate_type": "fixed" }
Response 200: { "status": "rate_selected", "monthly_payment": 1043.17 }

POST /api/mortgages/{id}/amortize
{ "amount": 5000 }
Response 200: { "remaining_principal": 195000, "status": "active" }

Leasing:

POST /api/leasing
{ "vehicle_id": "veh_1", "term_months": 36, "annual_km": 20000,
"down_payment": 5000, "insurance": "full" }
Response 201: { "leasing_id": "lsg_1", "status": "draft", "monthly": 429.00 }

POST /api/leasing/{id}/approve
Response 200: { "status": "active", "first_billing": "2026-12-01" }

Vault:

```
POST /api/vaults
 { "name": "Flat fund", "currency": "EUR",
  "members": ["usr_1","usr_2","usr_3"], "threshold": 2 }
 Response 201: { "vault_id": "vlt_1", "balance": 0.00 }

 POST /api/vaults/{id}/withdrawals
 { "amount": 100.00, "to_iban": "LT12...", "reason": "rent" }
 Response 201: { "withdrawal_id": "wdr_1", "status": "pending",
                 "approvals": 1, "required": 2 }

 POST /api/vaults/{id}/withdrawals/{wid}/approve
 { "decision": "approve" | "reject" }
 Response 200: { "status": "executed" | "pending" | "rejected" }
```

Card:

```
POST /api/cards
 { "account_id": "acc_1", "type": "virtual", "daily_limit": 500 }
 Response 201: { "card_id": "crd_1", "status": "active", "masked": "****4421" }
```

9.4 Events

Only one event is strictly needed: PaymentCompleted, to trigger round-up investment.

```
Event: PaymentCompleted
 Topic: payments.completed
 Producer: Payment module
 Consumer: Investment module

 {
  "transaction_id": "txn_1",
  "user_id": "usr_1",
  "amount": 2.40,
  "rounded_amount": 0.60,
  "currency": "EUR"
 }

 Rule: if rounded_amount > 0, create a micro-investment.
 If 0, ignore.
 Errors are logged and skipped (no retry logic needed).
```

EuriborUpdated is optional and can be replaced by a fixed rate for the academic demo.

9.5 Flows

```
Vault withdrawal:
 Member -> API: create withdrawal
 API -> Vault: check member and funds; reserve, notify signers
 API -> Member: 201 pending
 Member -> API: approve
 API -> Vault: check signer, register approval
 If threshold reached: execute payment, mark executed
 If expired (24h): mark expired, release funds, notify

 Micro-investment:
 Client -> Payment: card payment
 Payment: calculate round-up, save transaction, return OK
```

Payment -> EventBus: publish PaymentCompleted
EventBus -> Investment: deliver async
Investment: if round-up > 0, create investment order

10. Traceability Matrix

Requirement	Pattern(s)	Flow	Screen	Entity(ies)	Endpoint / Event
Registration and KYC	Facade	Onboarding	Register, KYC upload	User, KycRecord	POST /users/register
Multi-currency accounts	Composite	Create account	Accounts	Account, AccountMember	POST /accounts, GET /accounts/{id}
Transfers	Chain of Responsibility	Transfer	Transfer screen	Transfer, Account	POST /accounts/{id}/transfers
Smart mortgages	Strategy, State, Observer	Mortgage application	Mortgage simulator, status	Mortgage, MortgageRate, AmortizationSchedule	POST /mortgages, /rate-decision, /amortize
Car leasing	Builder, Facade	Leasing creation	Vehicle catalogue, configurator	LeasingContract, Vehicle, InsuranceOption	POST /leasing, /approve
Multi-signature vaults	Chain of Responsibility, State	Shared withdrawal	Vault dashboard, approvals	Vault, VaultMember, WithdrawalRequest, WithdrawalApproval	POST /vaults, /withdrawals, /approve
Micro-investments	Decorator, EDA	Payment round-up	Card payment, portfolio	Transaction, InvestmentOrder, InvestmentPortfolio	PaymentCompleted
Card management	State	Card issuance	Cards screen	Card, CardLimit	POST /cards
Investment analytics	Observer	Portfolio dashboard	Investment analytics	InvestmentPortfolio, PortfolioSnapshot	GET /investments/portfolio
Manual investments	Strategy	Manual investment	Investment marketplace	InvestmentOrder, Asset	POST /investments/orders
Insurance packages	Decorator	Leasing insurance	Leasing configurator	InsuranceOption, LeasingContract	Included in POST /leasing

11. Integration, Final QA and Evaluation

This section verifies that the document is complete, consistent and implementable by an external team without further explanations.

11.1 Implementer QA

Question	Answer	Owner
Withdrawal expires?	Becomes expired, funds released, members notified.	Vault
Payment fails after event?	Event only published after settlement; investment errors are logged and skipped.	Payment/Investment
Simultaneous approvals?	Database lock prevents double execution.	Vault
KYC incomplete?	Accounts, mortgages, leasing, cards, transfers return 422.	Identity
Euribor changes?	Optional; fixed rate is enough for demo.	Mortgage
Vault under 3 members?	Cannot create multi-signature vault.	Vault
Transfer exceeds funds?	422 insufficient funds; no reservation.	Account
Card frozen?	Payments rejected; account stays active.	Card
Leasing no insurance?	Insurance is mandatory; 400 if missing.	Leasing

11.2 Resolved gaps

Gap	Correction
Money format	Euros with 2 decimals
Vault approver	Active members only
Rejection effect	Reject cancels and releases funds
Duplicate events	Log and skip
Error format	Standard HTTP codes
2FA/SCA	Removed for academic scope

11.3 Glossary

Term	Definition
KYC	Identity verification
AML	Anti-money-laundering controls
Vault	Shared account with multi-signature rules
Threshold	Minimum approvals needed
Round-up	Difference to next whole euro
Event Bus	Async publish/consume channel
RPO	Max data loss
RTO	Max restoration time

11.4 Technical decisions

Decision	Justification
Amounts in euros (2 decimals)	Simpler than integer cents for academic scope
Simple token auth	Enough for a prototype; no 2FA needed
Async round-ups	Payments do not depend on investments
Vault min 3 members	Makes multi-signature meaningful
24h withdrawal expiry	Prevents indefinite fund reservation
Standard HTTP errors	Easier to implement and understand
Designed specified 2FA/SCA but API prototype allows simple token auth	To facilitate student implementation

11.5 Risks

Risk	Mitigation
Double withdrawal	Locking + executed state
Lost event	Log error (no retry needed for prototype)
No KYC	Financial products blocked
Investment failure	Fault isolation; payments continue
Data breach	HTTPS + basic validation